

Exam

Name _____

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 1) Which is NOT considered to be a primary function of the respiratory system? 1) _____
 A) protection against pathogens
 B) vocalization
 C) regulation of pH balance
 D) regulation of water balance
 E) gas exchange

- 2) Ventilation is also known as 2) _____
 A) expiration.
 B) breathing.
 C) blowing.
 D) air conduction.
 E) inspiration.

- 3) The upper respiratory tract includes all EXCEPT which of the following? 3) _____
 A) mouth B) nasal cavity C) lungs D) larynx

- 4) The lower respiratory tract includes 4) _____
 A) all of the bronchial branches and the lungs.
 B) all of the bronchial branches, the lungs, and the trachea.
 C) all of the bronchial branches.
 D) only the lungs.
 E) only the trachea.

- 5) Pulmonary ventilation refers to the 5) _____
 A) movement of dissolved gases from the blood to the interstitial space.
 B) movement of dissolved gases from the interstitial space to the cells.
 C) movement of dissolved gases from the alveoli to the blood.
 D) movement of air into and out of the lungs.
 E) utilization of oxygen.

- 6) Alveolar ventilation refers to the 6) _____
 A) movement of air into and out of the alveoli.
 B) movement of dissolved gases from the alveoli to the blood.
 C) movement of air into and out of the lungs.
 D) movement of dissolved gases from the blood to the alveoli.
 E) utilization of oxygen by alveolar cells to support metabolism.

- 7) The sites of gas exchange within the lungs are 7) _____
 A) bronchioles.
 B) alveolar ducts.
 C) alveoli.
 D) terminal sacs.
 E) pleural spaces.



- 8) Place the following structures of the respiratory tree in the order in which air passes through them. 8) _____
1. secondary bronchi
 2. bronchioles
 3. primary bronchi
 4. alveoli
 5. terminal bronchioles
- A) 4, 1, 2, 3, 5 B) 3, 1, 2, 5, 4 C) 1, 3, 2, 5, 4 D) 1, 3, 5, 2, 4 E) 3, 1, 5, 2, 4
- 9) The airway between the larynx and the primary bronchi is the 9) _____
- A) trachea.
 - B) laryngeal duct.
 - C) alveolar duct.
 - D) bronchiole.
 - E) pharynx.
- 10) The lungs are surrounded by _____ membranes. 10) _____
- A) pericardial
 - B) pleural
 - C) pulmonary
 - D) costal
 - E) thoracic
- 11) The lungs are located in the _____ cavity. 11) _____
- A) pleural
 - B) thoracic
 - C) costal
 - D) pulmonary
 - E) pericardial
- 12) Pressure and volume of gas in a container are related to temperature and number of gas molecules. 12) _____
- This is known as _____ law.
- A) Ohm's
 - B) Dalton's
 - C) the ideal gas
 - D) Boyle's
 - E) Henry's
- 13) Type II alveolar cells 13) _____
- A) allow rapid diffusion of gases through their thin membranes, secrete a chemical known as surfactant, and are phagocytic.
 - B) are phagocytic.
 - C) secrete a chemical known as surfactant.
 - D) allow rapid diffusion of gases through their thin membranes.
 - E) None of the statements are true.



- 14) Type I alveolar cells 14) _____
 A) are phagocytic.
 B) allow rapid diffusion of gases through their thin membranes.
 C) allow rapid diffusion of gases through their thin membranes, secrete a chemical known as surfactant, and are phagocytic.
 D) secrete a chemical known as surfactant.
 E) None of the statements are true.
- 15) Which feature of the alveoli allows for the ease of diffusion of gases? 15) _____
 A) Elastin fibers allow the alveoli to stretch thin enough for diffusion to occur.
 B) Type II alveolar cells secrete surfactant.
 C) They are made of a single layer of epithelium.
 D) They are made of a single layer of simple squamous epithelium and elastin fibers allow the alveoli to stretch thin enough for diffusion to occur.
- 16) Surfactant 16) _____
 A) is not found in healthy lung tissue.
 B) protects the surface of the lungs.
 C) phagocytizes small particulate matter.
 D) helps prevent the alveoli from collapsing.
 E) replaces mucus in the alveoli.
- 17) The common passageway shared by the respiratory and digestive systems is the 17) _____
 A) pharynx.
 B) esophagus.
 C) vestibule.
 D) glottis.
 E) larynx.
- 18) When the diaphragm and external intercostal muscles contract, 18) _____
 A) the volume of the thorax increases.
 B) the volume of the thorax decreases.
 C) expiration occurs.
 D) the lungs collapse.
 E) the volume of the lungs decreases.
- 19) Dalton's law states that 19) _____
 A) the volume of gas that will dissolve in a solvent is proportional to the solubility of the gas and the gas pressure.
 B) in a mixture of gases like air, the total pressure is the sum of the individual partial pressures of the gases in the mixture.
 C) gas volume and temperature are directly proportional.
 D) gas volume and pressure are inversely proportional.
 E) None of the answers are correct.
- 20) Air moves into the lungs because 20) _____
 A) the thorax is muscular.
 B) contraction of the diaphragm decreases the volume of the pleural cavity.
 C) the volume of the lungs decreases with inspiration.
 D) the gas pressure in the lungs is less than outside pressure.
 E) All of the answers are correct.

